

Written by: Dr. Farkhana Muchtar

LAB 0: INSTALLATION EXERCISE

Installing and Running Linux OS in Windows 10 Using Oracle VirtualBox

1 Installation Steps

1.1 Windows OS Command Line Environment

We will use PowerShell 7 (an open-source tool) and Windows Terminal to improve the Windows command line, making it easier to use and more visually appealing than the default options.

Windows 10/11 includes two command line tools by default: Command Prompt (an older tool from DOS, Windows 95, and Windows NT) and Windows PowerShell (more powerful but not very attractive).

1.1.1 Installing Open Source PowerShell (PowerShell 7)

Step 1: Download PowerShell 7 Installer from this link

<https://github.com/PowerShell/PowerShell/releases/tag/v7.5.0>

Step 2: Scroll down until the Assets section. Choose **PowerShell-7.5.0-win-x64.msi** file.

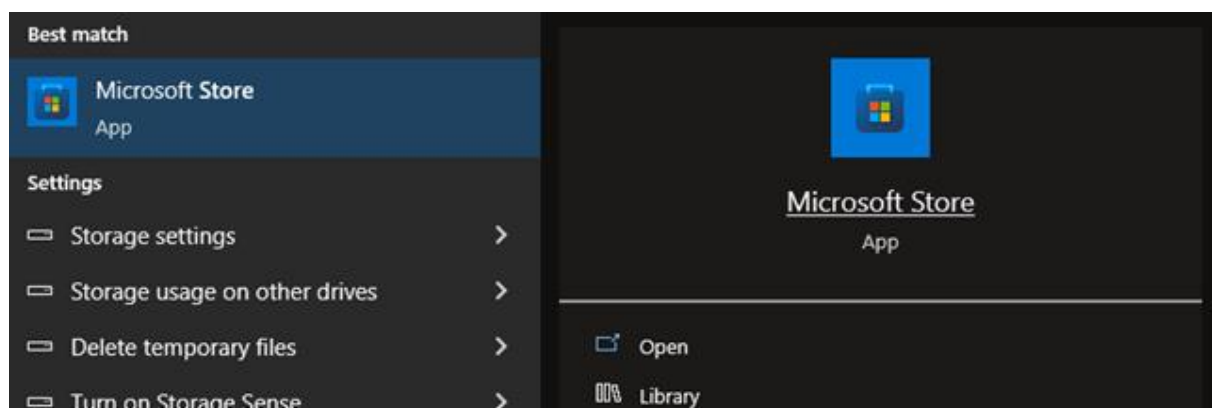
Step 3: After download, simply double click the installer file

1.1.2 Installing Windows Terminal

Step 1: Search **Store** at Windows Search



Step 2: Click **Microsoft Store**

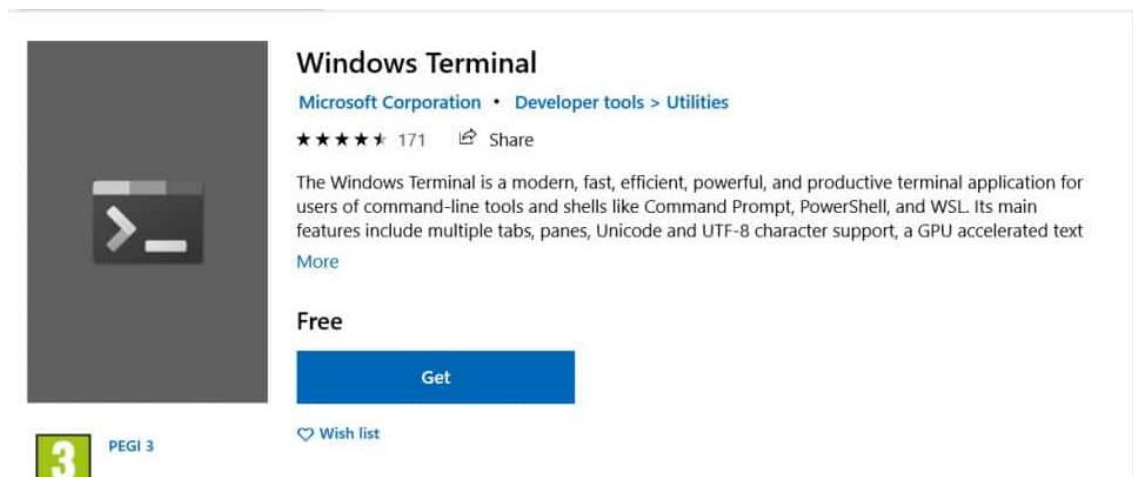


Step 3: In Microsoft Store, search for **Windows Terminal**

Step 4: Simply click **Get** button to install **Windows Terminal**.

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If you have trouble installing Windows Terminal from the Microsoft Store, try this alternative method instead.

Step 1: Go to <https://conemu.github.io/>.

Step 2: Find download button and click the button to go to the github download page.

Step 3: Locate the correct setup file and click it to download it to your computer.

Step 4: Click the downloaded file to install **conemu**.



Conemu might not as user friendly as **Windows Terminal** but it has more features that you can explore.

* For Mac OS user, it is enough to use default **Terminal** app in your Mac OS. Just search **Terminal** in Finder to start using **Terminal** in your Mac OS.

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1.2 Installing Virtualbox as Virtualization Solution

VirtualBox is a free Type 2 hypervisor that lets you run virtual machines quickly and reliably, depending on your computer's hardware. Even though it is entirely free, VirtualBox still provides features that many other hypervisors either do not have or need a premium subscription to use.

Step 1: Go to the Oracle VirtualBox download page using the below link.

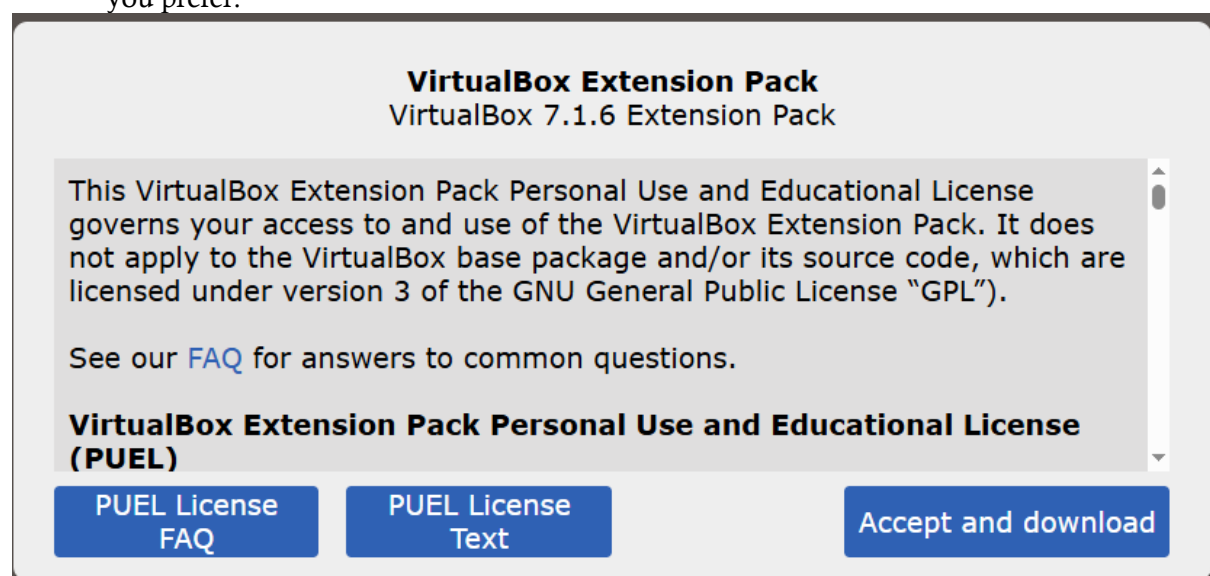
<https://www.virtualbox.org/wiki/Downloads>

Step 2: Click on the Windows hosts link. Save the installer file anywhere, you prefer.



Step 3: Navigate to the installer file and double-click on it for the installation process. Stick with the default settings during the installation process.

Step 4: At the same website page, under the 'VirtualBox Extension Pack' section, click the 'Accept and download' button to download the latest pack. Save the installer file anywhere, you prefer.



Step 5: Navigate to the installer file and double-click on it for the installation process. Now the Virtualbox already installed in your computer/laptop.

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1.3 Steps to Install Ubuntu 24.04 LTS Minimal Cloud Image on Virtual Box

Step 1: Download Ubuntu 24.04 LTS Cloud Image

Since we're installing Ubuntu 24.04 Noble Numbat using VirtualBox, you can download a ready-made file (called an appliance) from the Ubuntu 24.04 CD Image page.

Visit this link

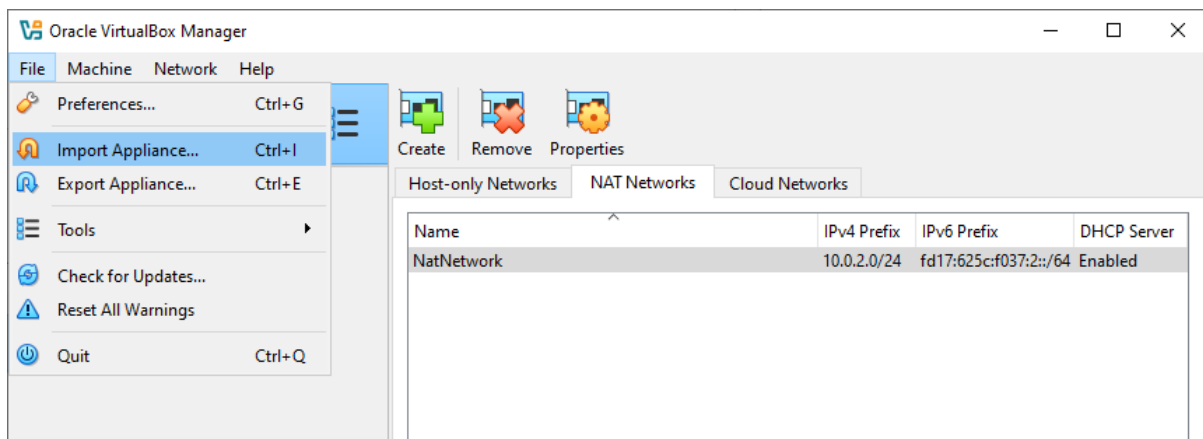
<https://cloud-images.ubuntu.com/noble/current/>

and on the page, download **noble-server-cloudimg-amd64.ova** file.

	noble-server-cloudimg-amd64-azure.vhd.tar.gz	2025-04-03 04:22	557M	File system image and Ke
	noble-server-cloudimg-amd64-lxd.tar.xz	2025-04-03 13:20	412	Ubuntu Server 24.04 LTS
	noble-server-cloudimg-amd64-root.manifest	2025-04-03 13:18	18K	Package manifest file
	noble-server-cloudimg-amd64-root.tar.xz	2025-04-03 13:18	209M	Ubuntu Server 24.04 LTS
	noble-server-cloudimg-amd64.daily.20250327.20250403.image_changelog.json	2025-04-03 13:28	234K	Image Changelog
	noble-server-cloudimg-amd64.img	2025-04-03 13:17	583M	QCow2 UEFI/GPT Bootable
	noble-server-cloudimg-amd64.manifest	2025-04-03 13:17	20K	Package manifest file
	noble-server-cloudimg-amd64.ova	2025-04-03 13:18	554M	VMware/Virtualbox OVA

Step 2: Import Appliance in VirtualBox

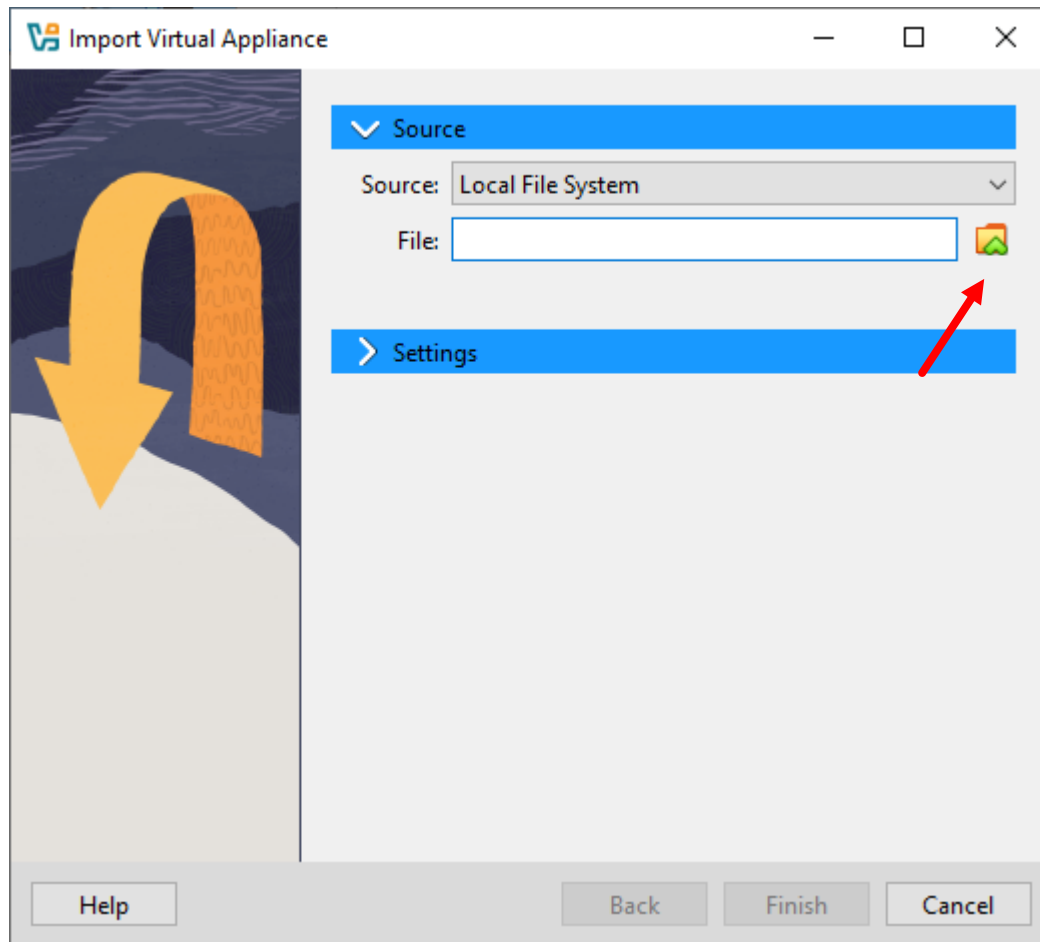
Open VirtualBox on your computer. Then, go to the **File** menu and choose '**Import Appliance...**'.



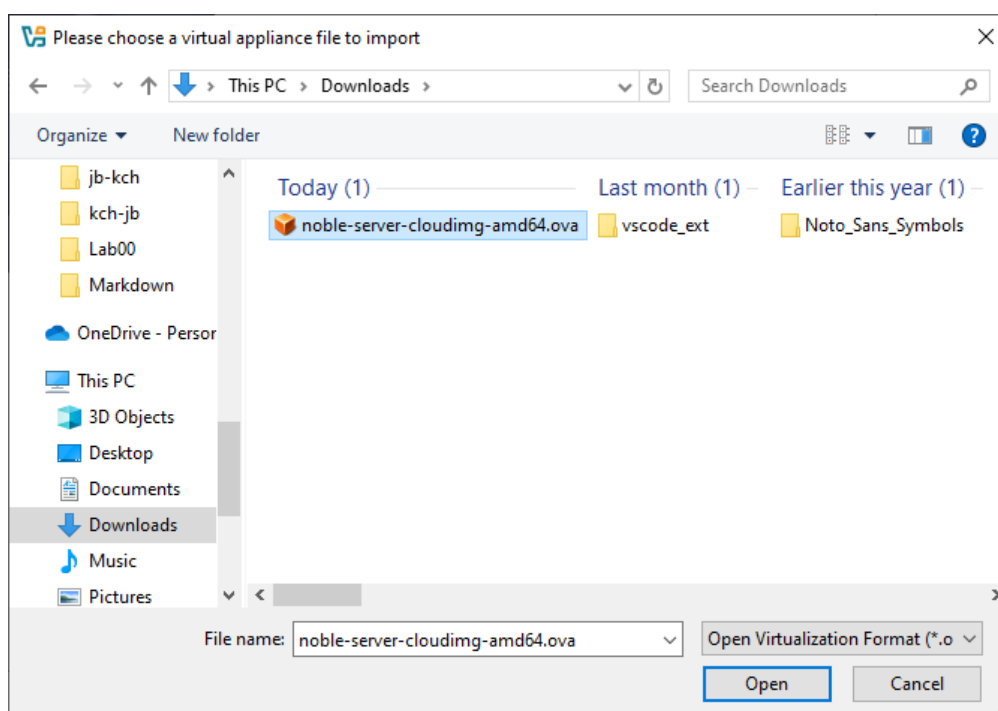
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Click the folder icon

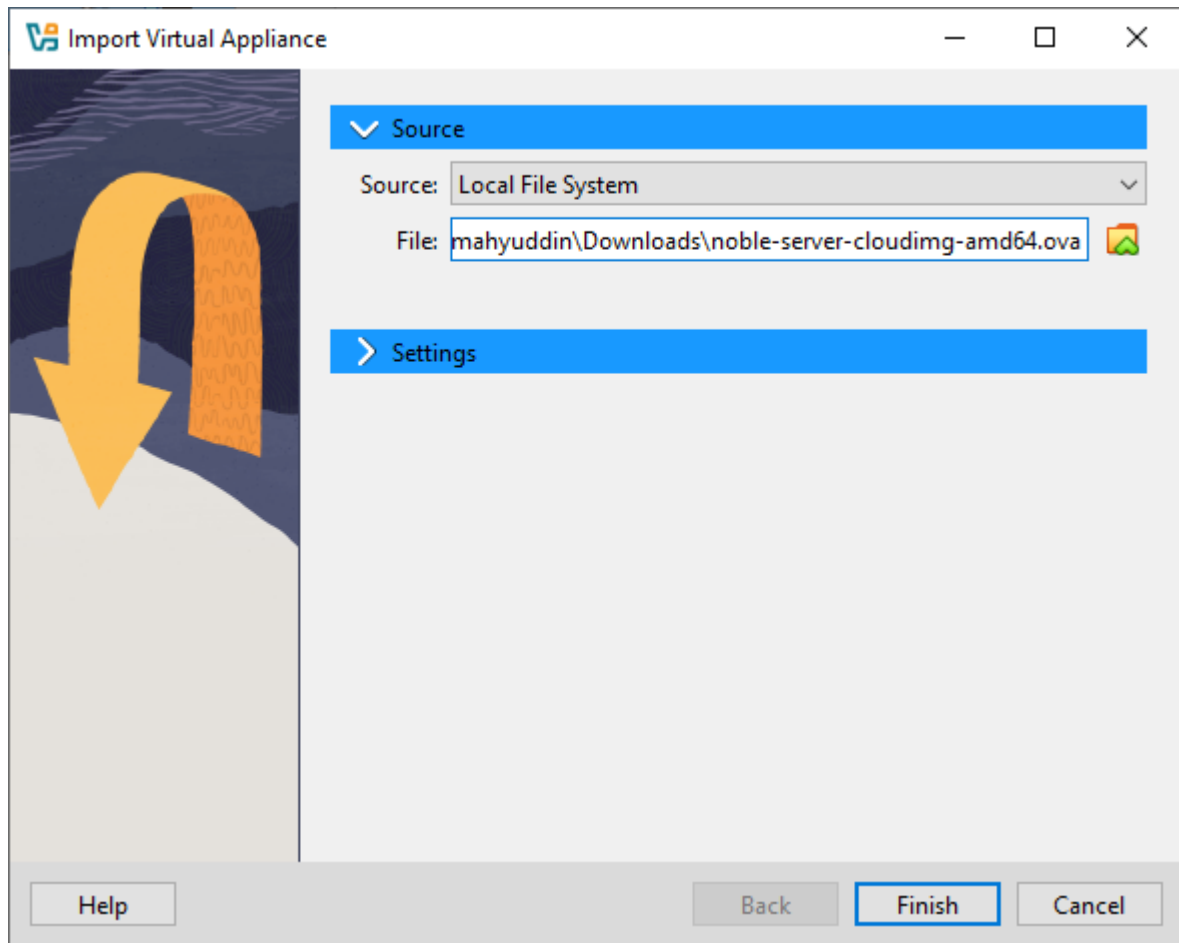


and locate the Ubuntu 24.04 Noble Numbat OVA file you downloaded. Then, click the 'Open' button.



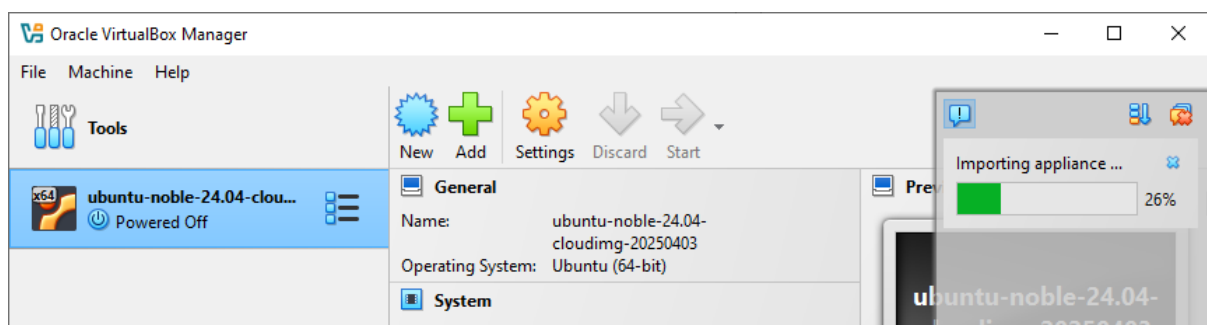
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After that, simply hit the **Finish** button.

Virtualbox will import the appliance image and creating a new virtual machine based on that image.

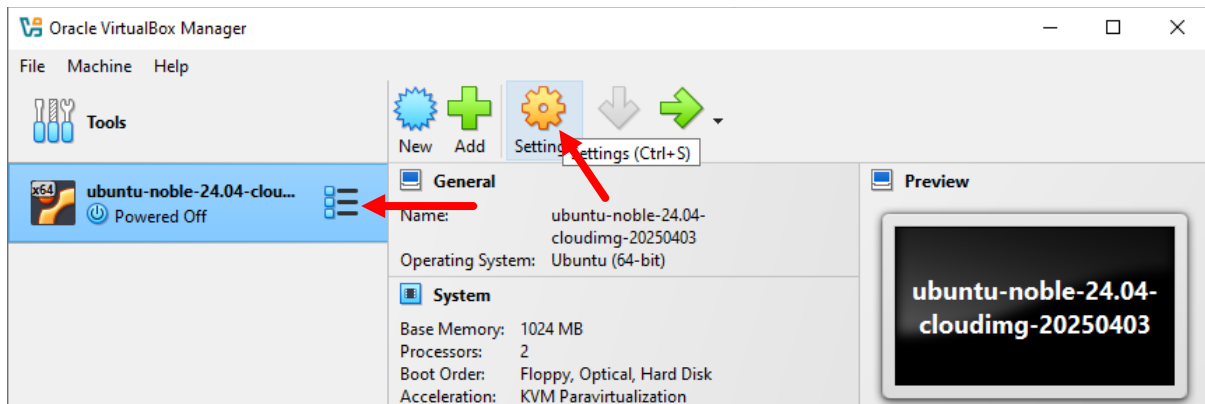


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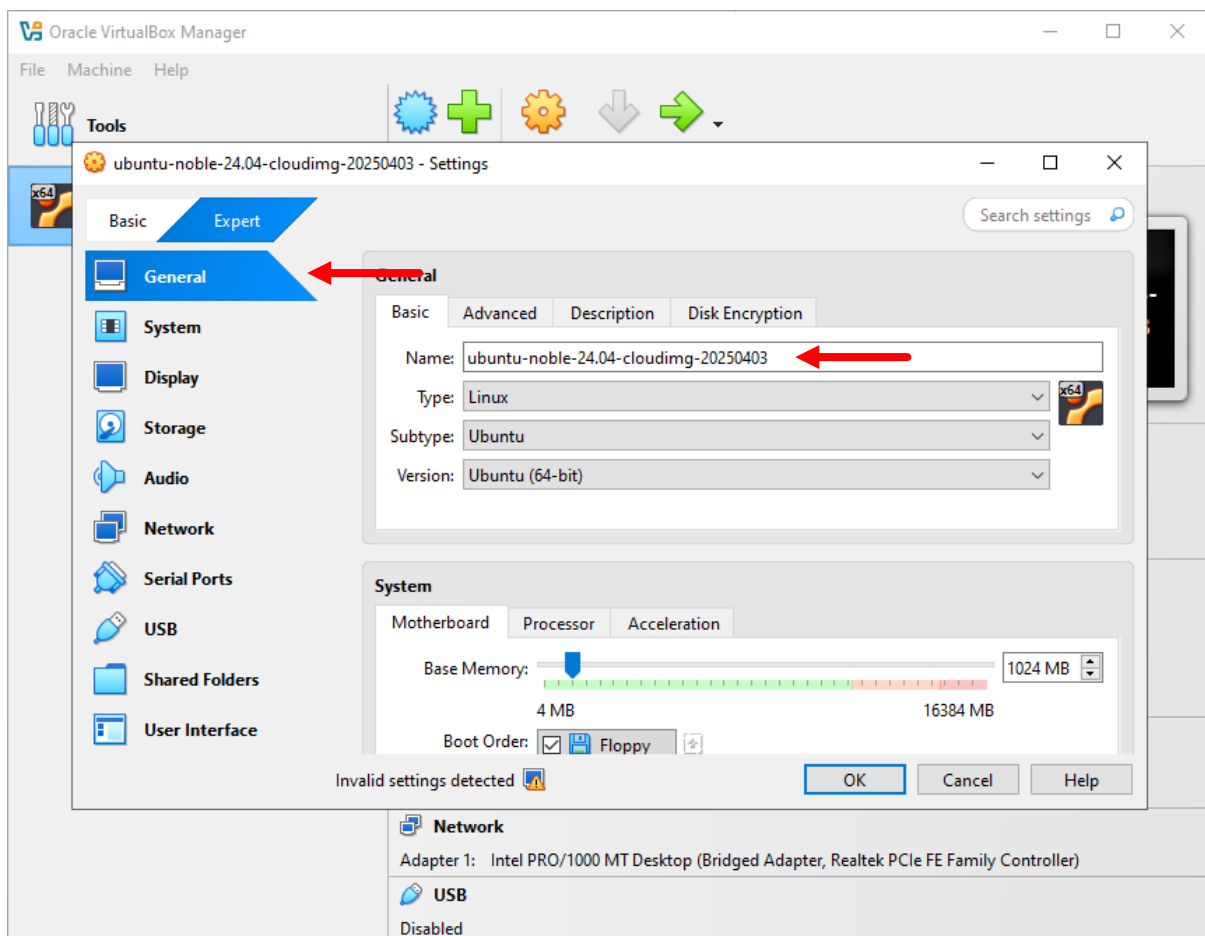
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Step 3: Setting up the virtual machine

Select the virtual machine and click Setting icon to change the setting of the virtual machine.



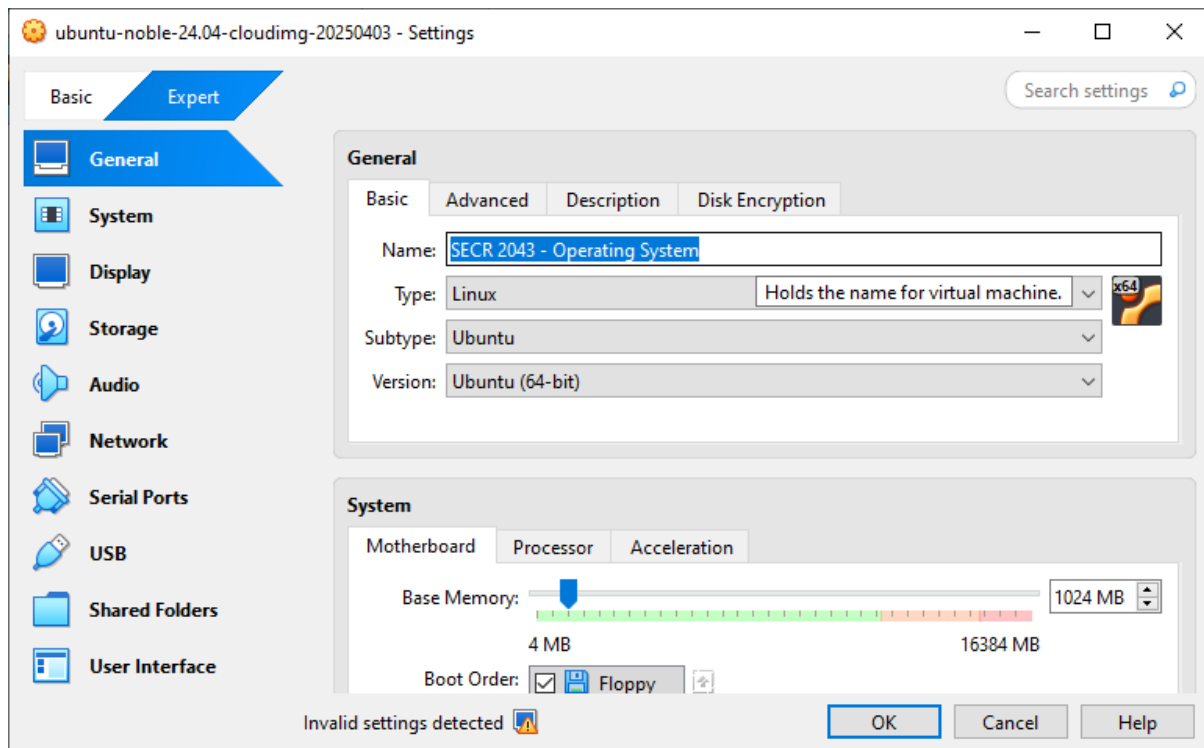
First, you need to set a name for your new virtual machine. By default, you will be to setting up **General** configuration of the virtual machine.



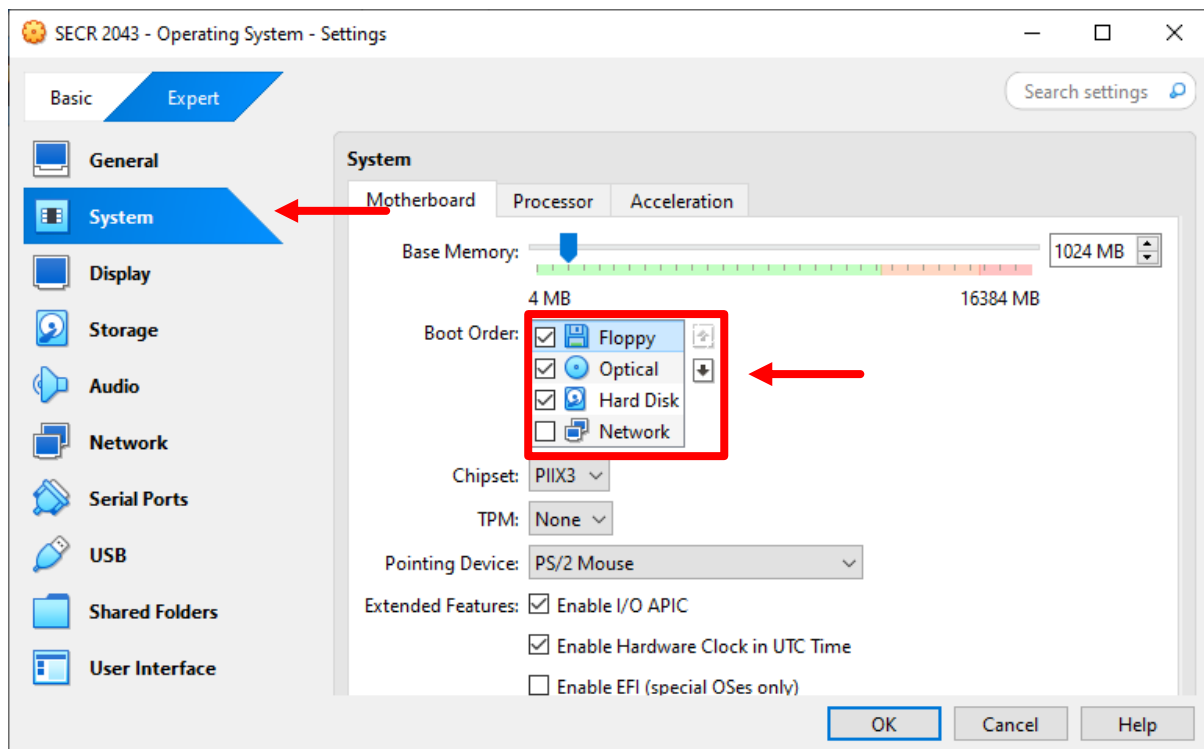
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In these settings, we'll simply give our virtual machine a more descriptive name. For example, **SECR 2043 – Operating System**.



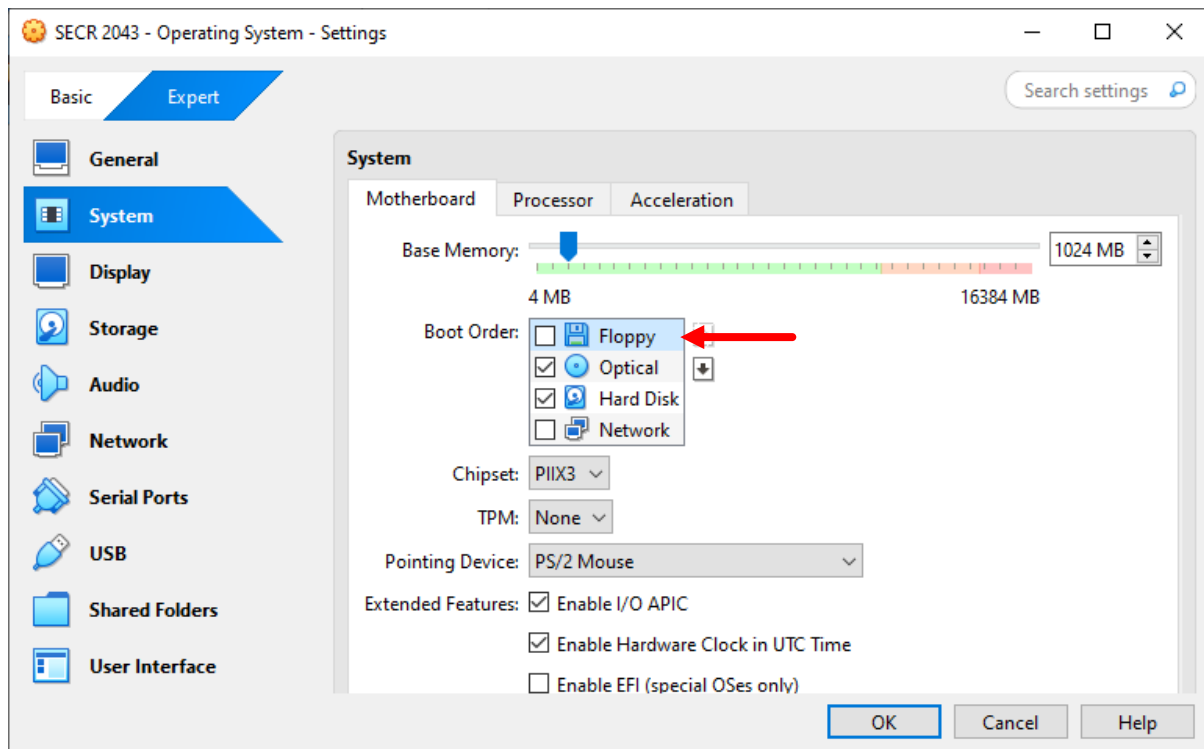
After that, we will do some **System** setting, specifically on **Boot Order** of our virtual machine.



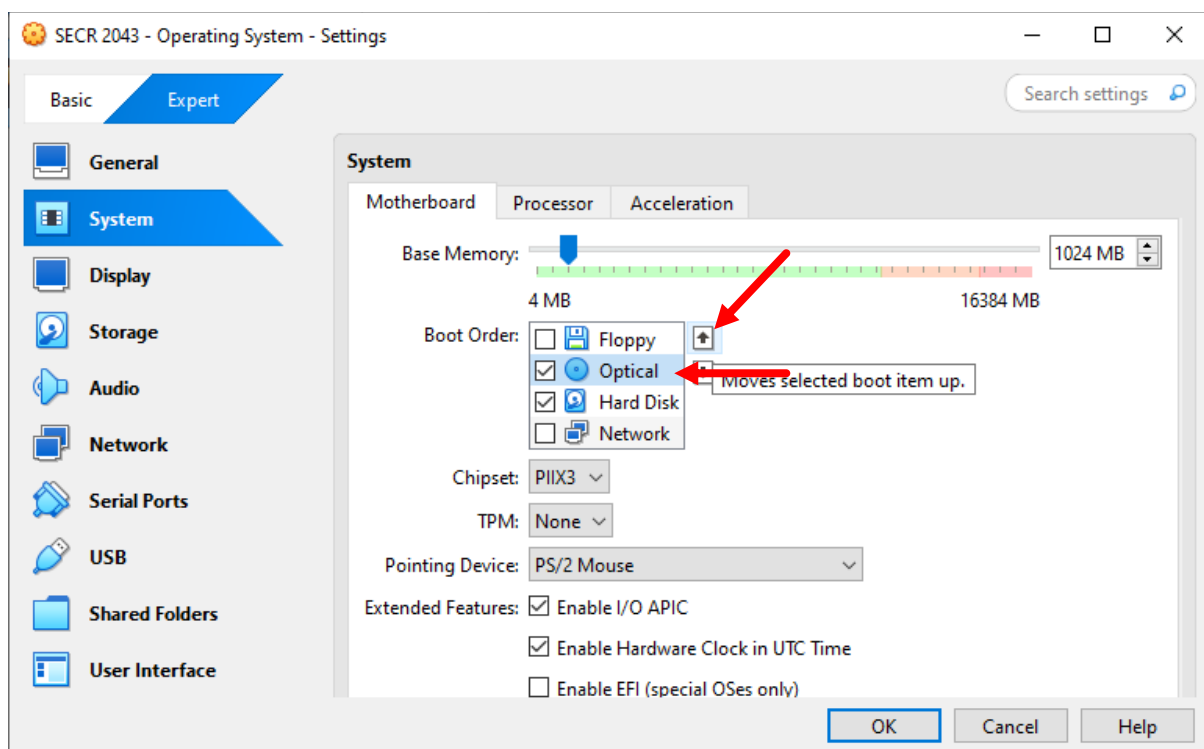
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Turn off the Floppy disk boot option by unchecking the **Floppy** box, as shown below.



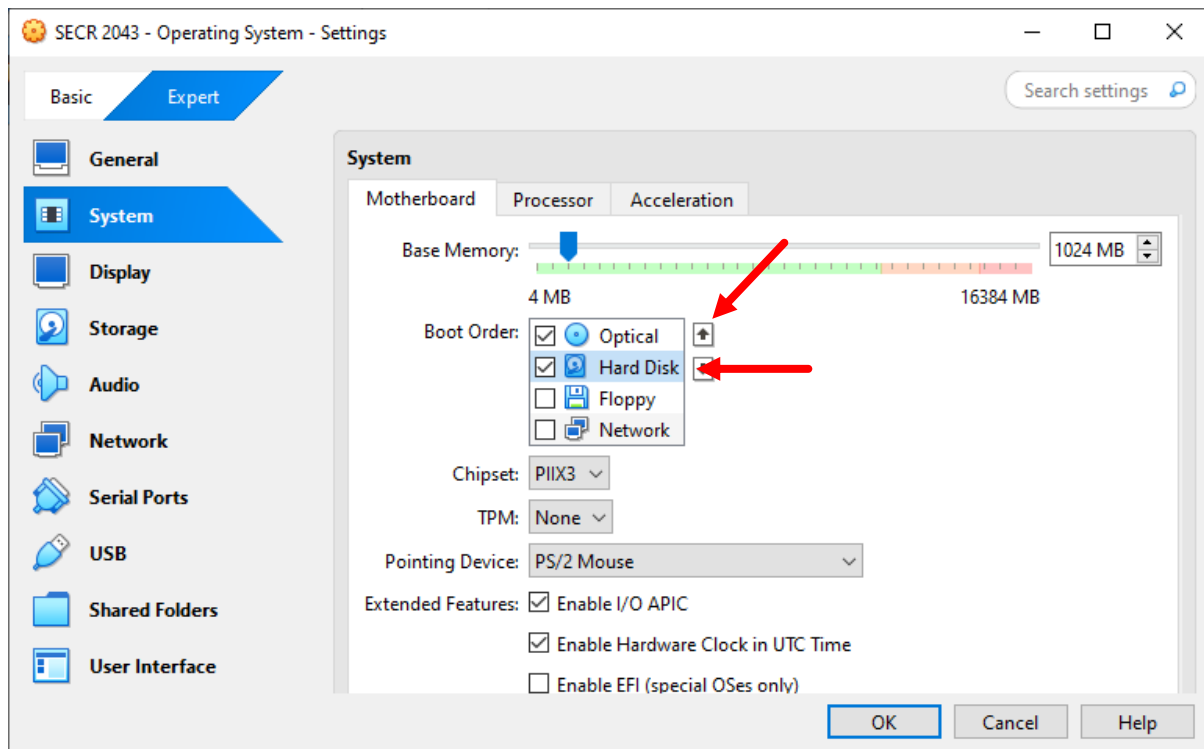
Next, select the **Optical** device and move it to the top of the boot order.



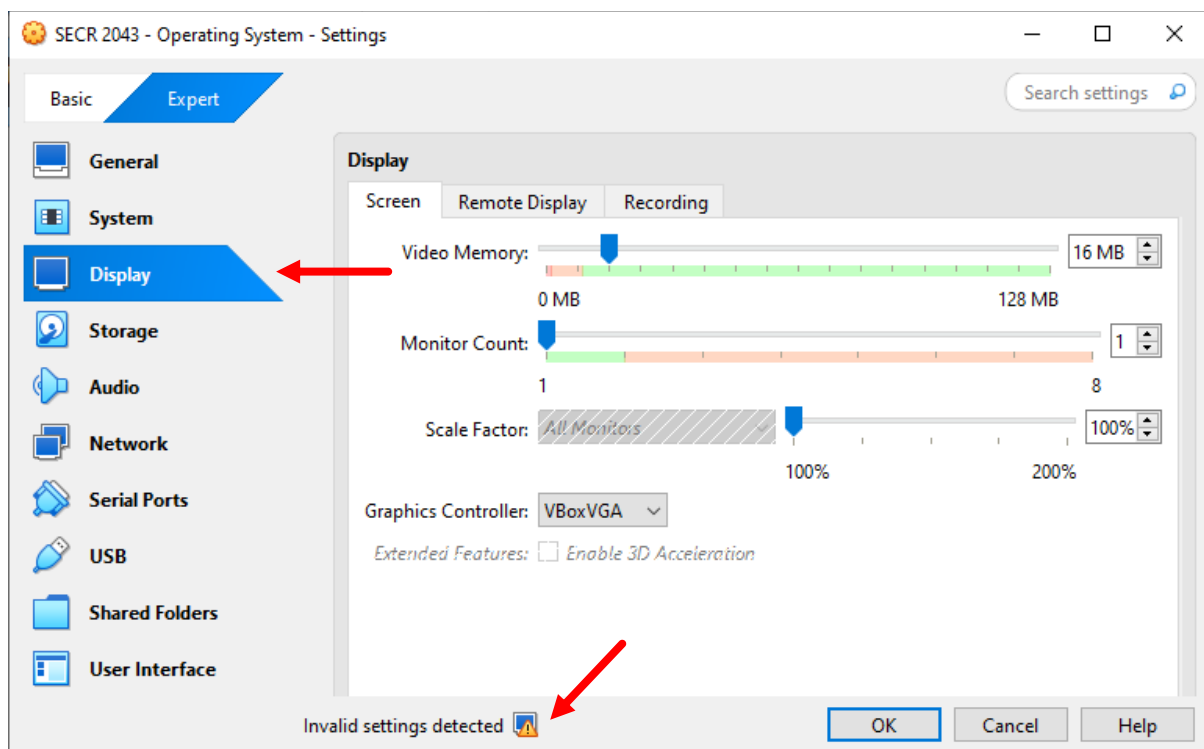
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Do the same to **Hard Disk** as second boot device in **Boot Order**.



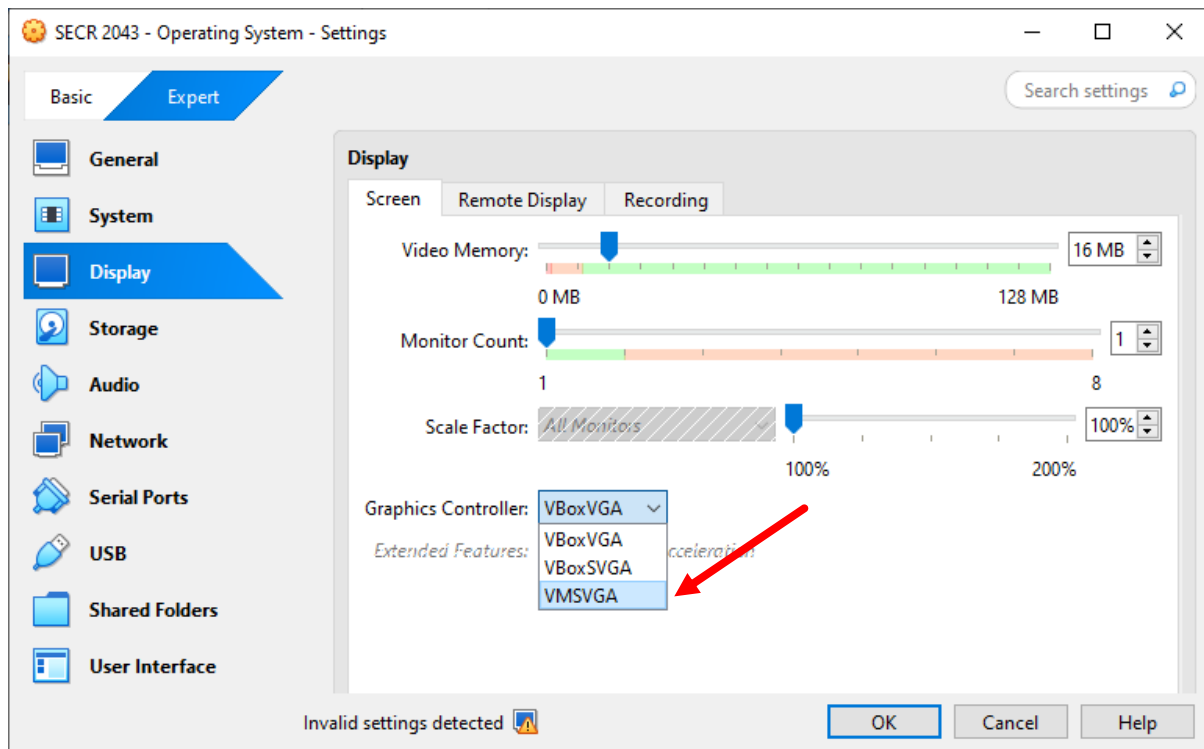
Now we move to Display setting of our virtual machine to fix some setting error as shown in the screenshot below.



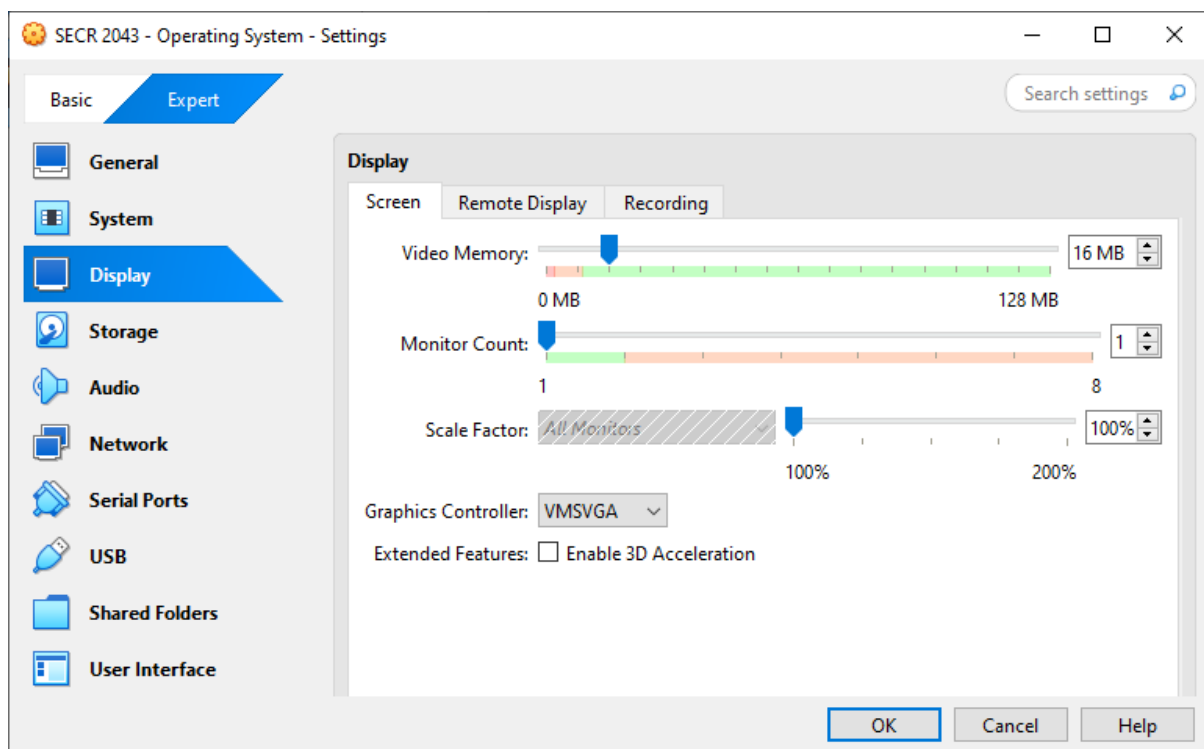
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Change the **Graphics Controller** from **VBoxVGA** to **VMSVGA** like below.



Now, we don't have invalid setting warning message anymore.



Don't forget to click OK button to confirm the configuration changes.

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Step 4: seed.iso for appliance setup

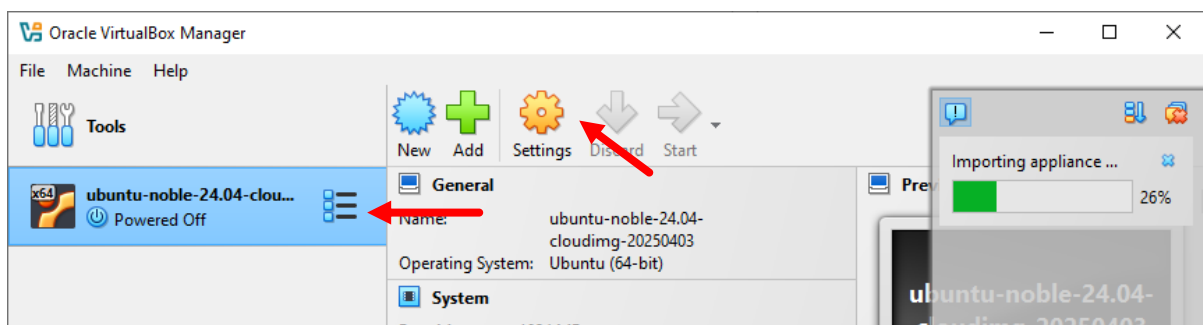
By default, you want to be able to log in to this minimal Ubuntu server image but there is no existing user with password.

To fix this, we'll use a **seed.iso** file to set up a username and password. We'll also use the **seed.iso** file to pre-set the network, hostname, and basic SSH server settings so you can remotely access the virtual machine.

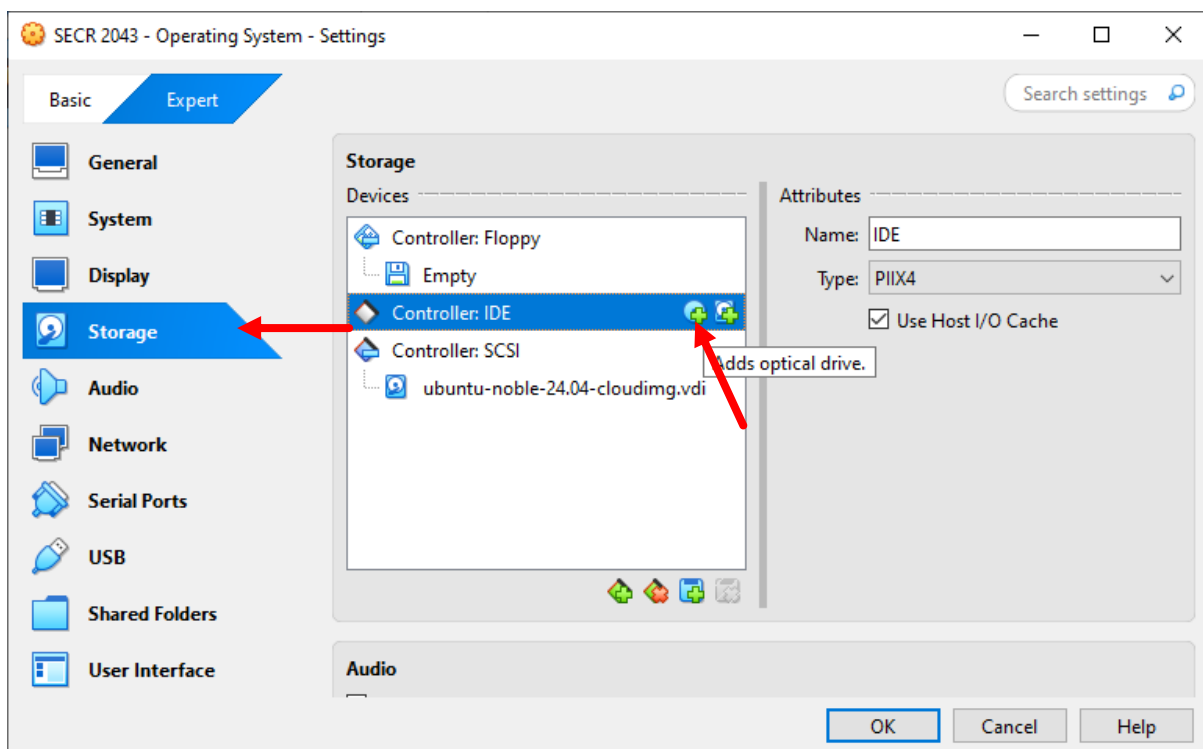
First of all, you need to download **seed.iso** that I created for this lab exercise from the link below and save it at any location that you desired such as Downloads folder.

<https://drive.google.com/uc?export=download&id=1wfrUCPkuiTzDySispJ898b8yzpFFKg4D>

Now, select the **imported** Virtual machine (created using the Cloud image OVA) and hit the **Settings** button.



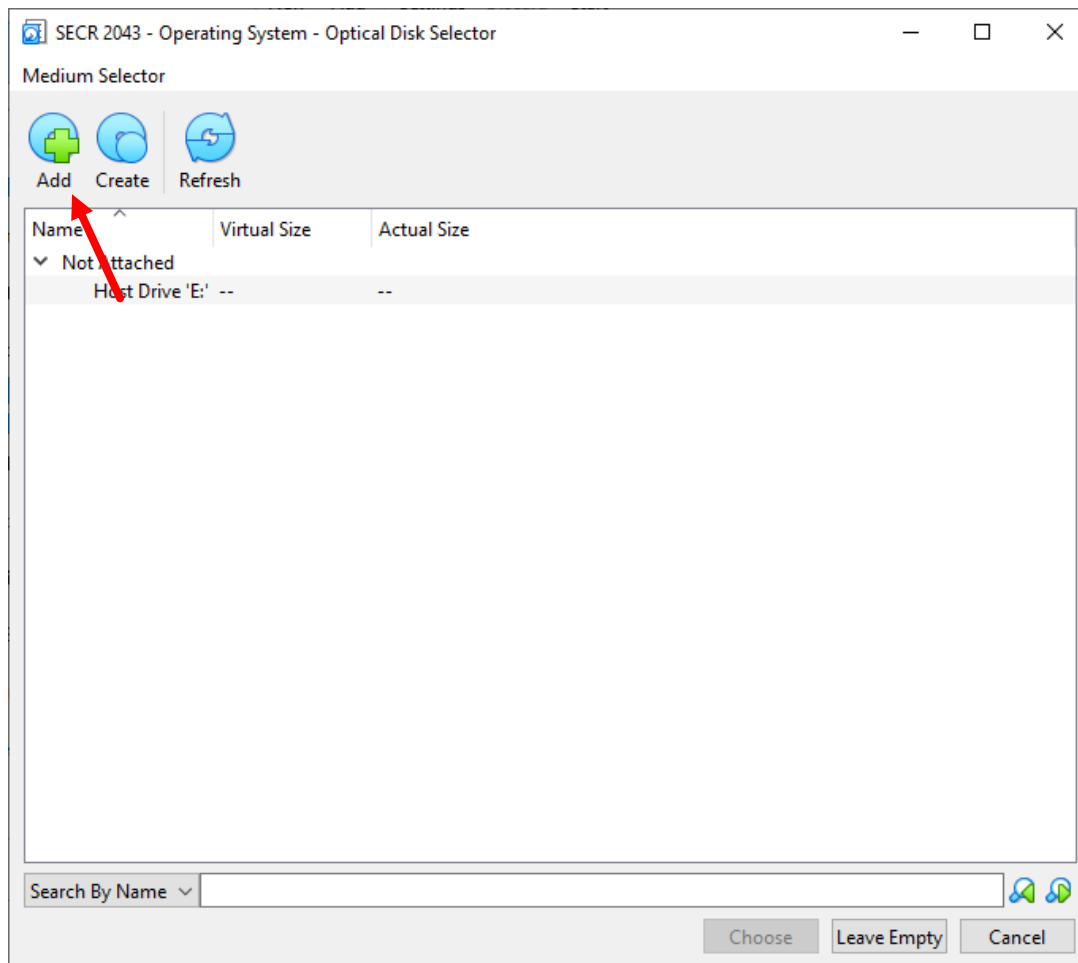
After that, click to **Storage**, and then click **add optical drive** icon at **Controller IDE** to add **seed.iso** as new virtual cd to our virtual machine.



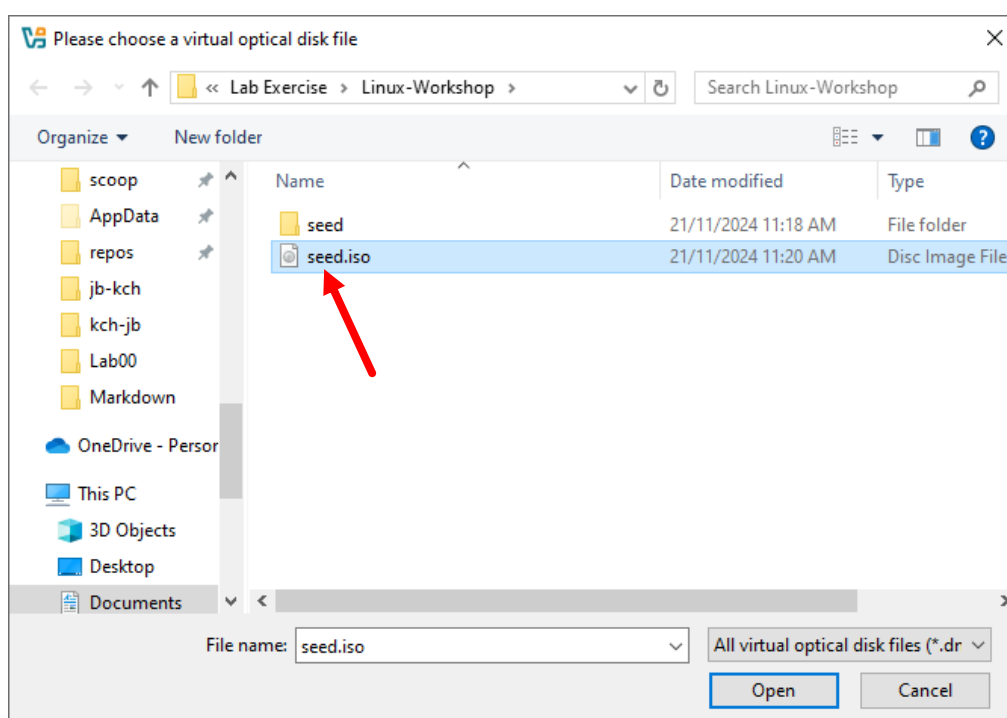
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To do that, click on **Add** button and file explorer will open.



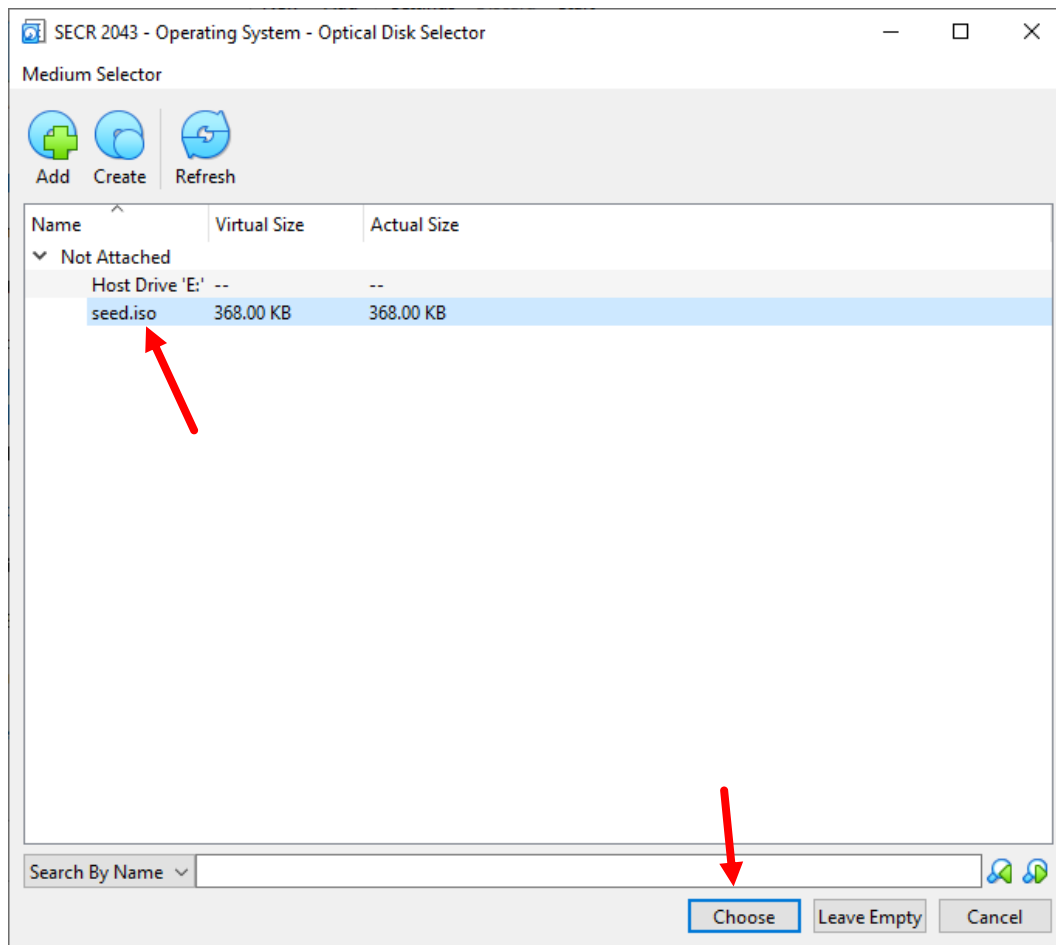
Now, select the **seed.iso** file which you have downloaded.



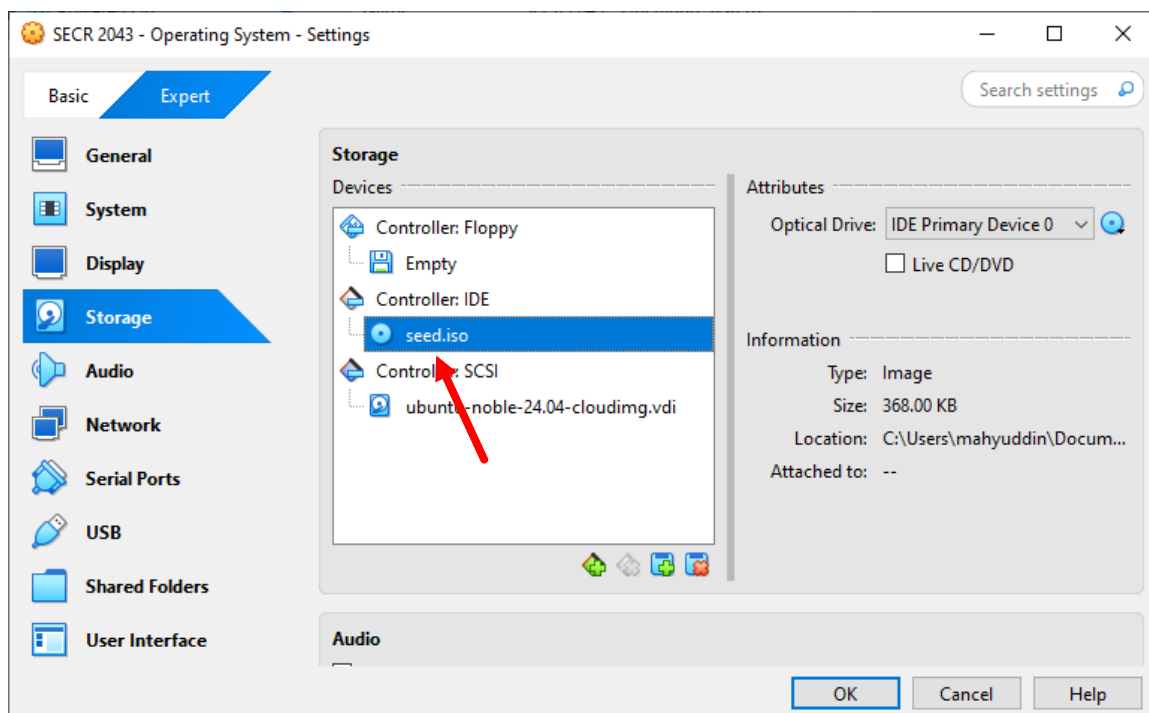
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After that, it will show in the **File selection** area, **again select** it and hit the **Choose** button.



Now, you have **seed.iso** as bootable virtual optical disk (CD) for you virtual machine



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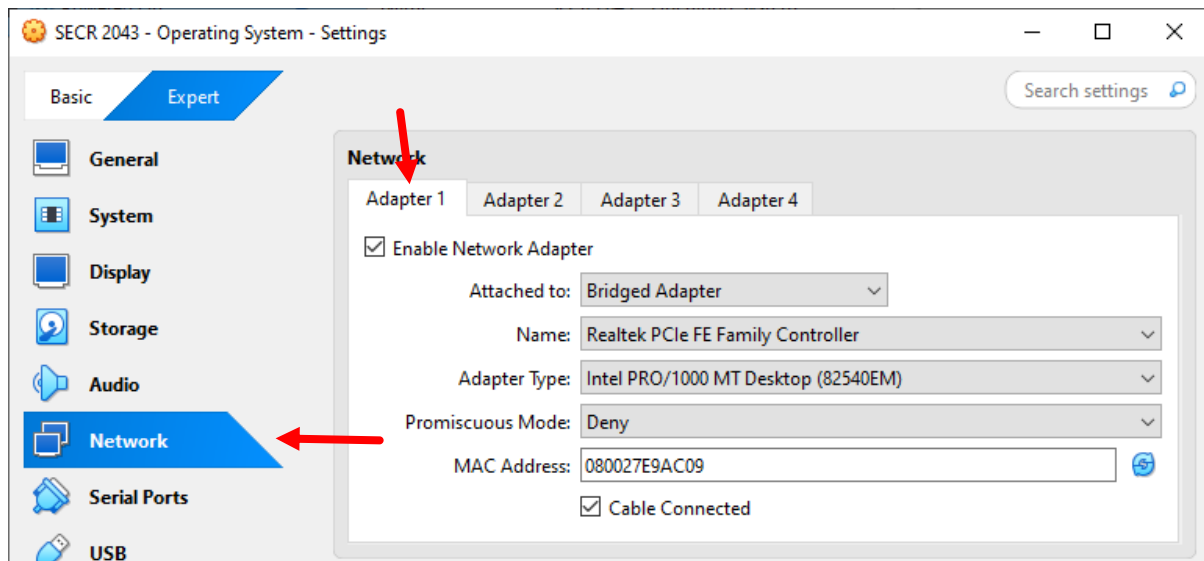
Step 5: Network Adapter Setup

We need to set up two network adapters for this virtual machine.

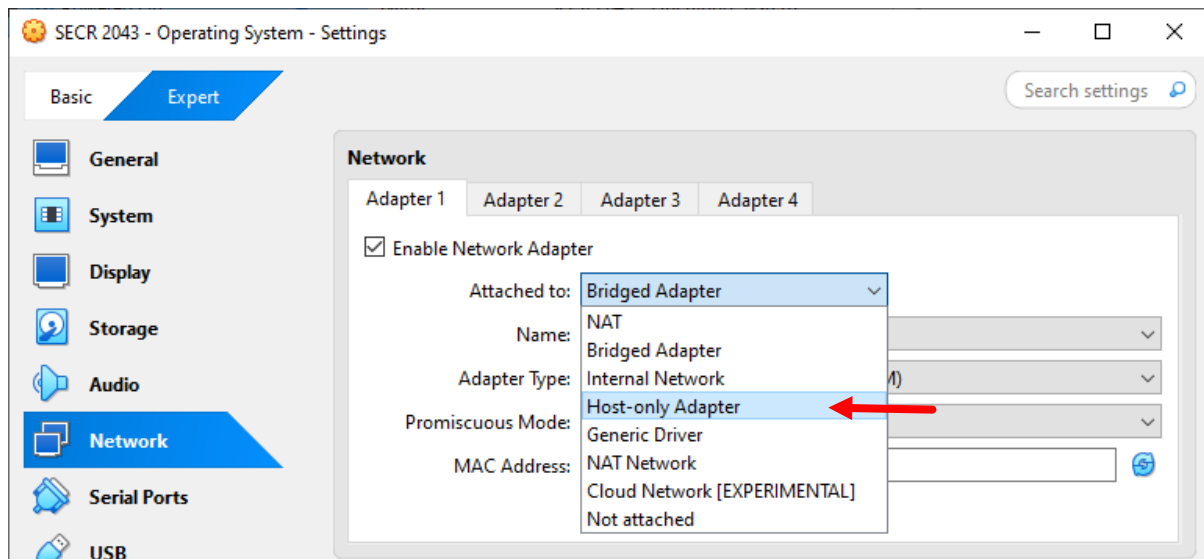
1. **Adapter 1** lets us connect to the virtual machine remotely using SSH.
2. **Adapter 2** is for this virtual machine to have Internet connection.

To achieve that, we set **Adapter 1** using **Host-Only Adapter**

Originally, Adapter 1 is attached to **Bridged Adapter**.



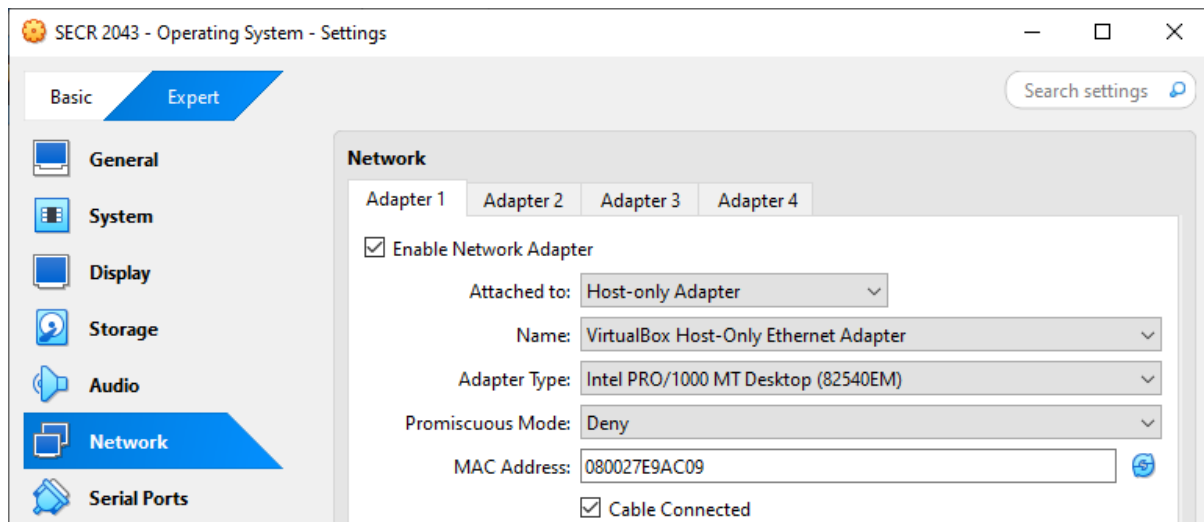
Change the **Bridged Adapter** to **Host-only Adapter** for **Adapter 1**.



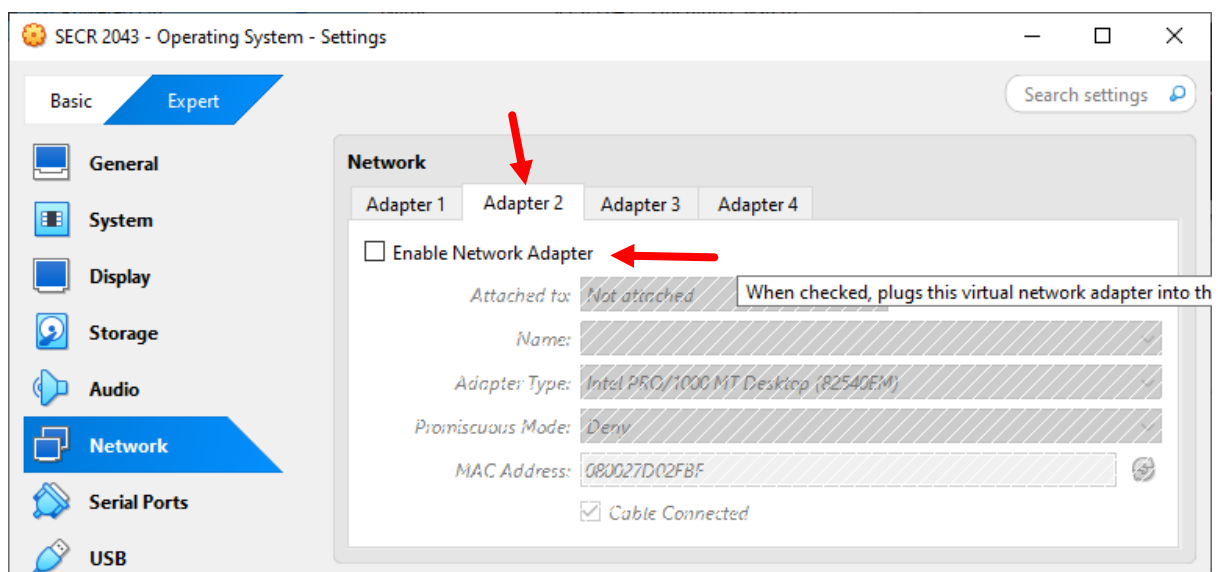
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Now we have **Adapter 1** that is attached to **Host-only Adapter**. This adapter is dedicated to allow host machine (your laptop) to access the virtual machine remotely.

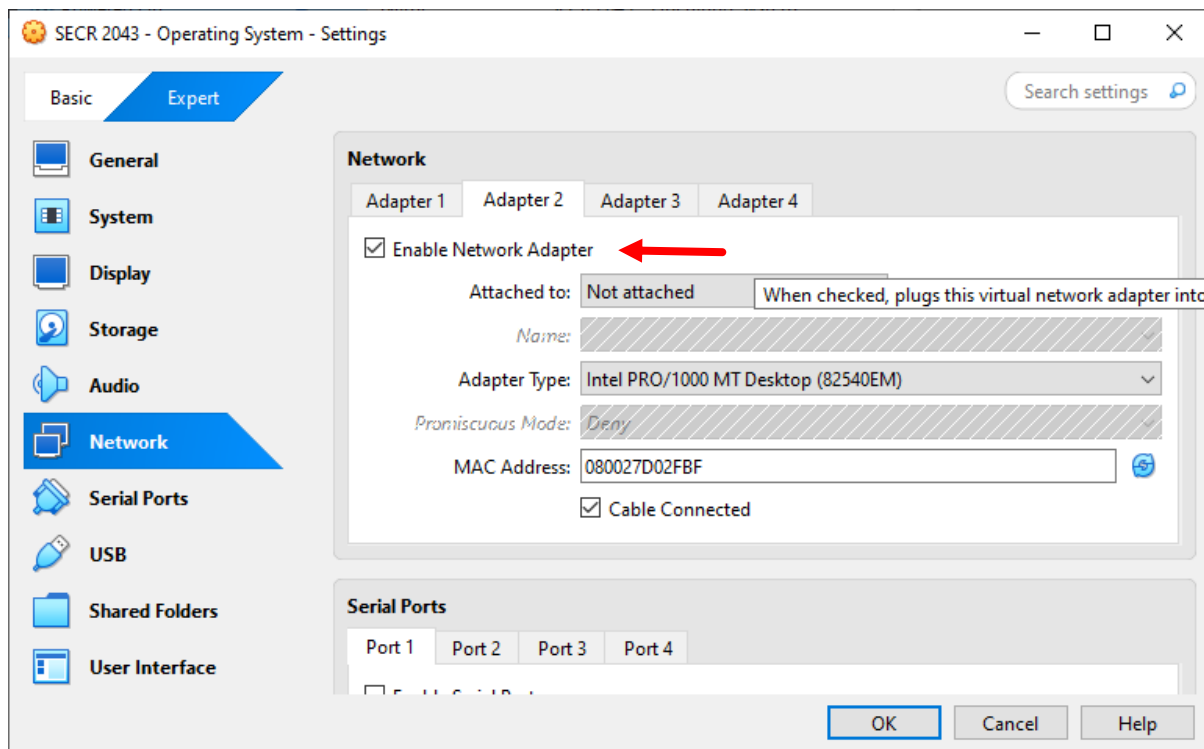


Next, we move to the **Adapter 2**. By default, this adapter is **disabled**. Just enable it by checking the **Enable Network Adapter**.

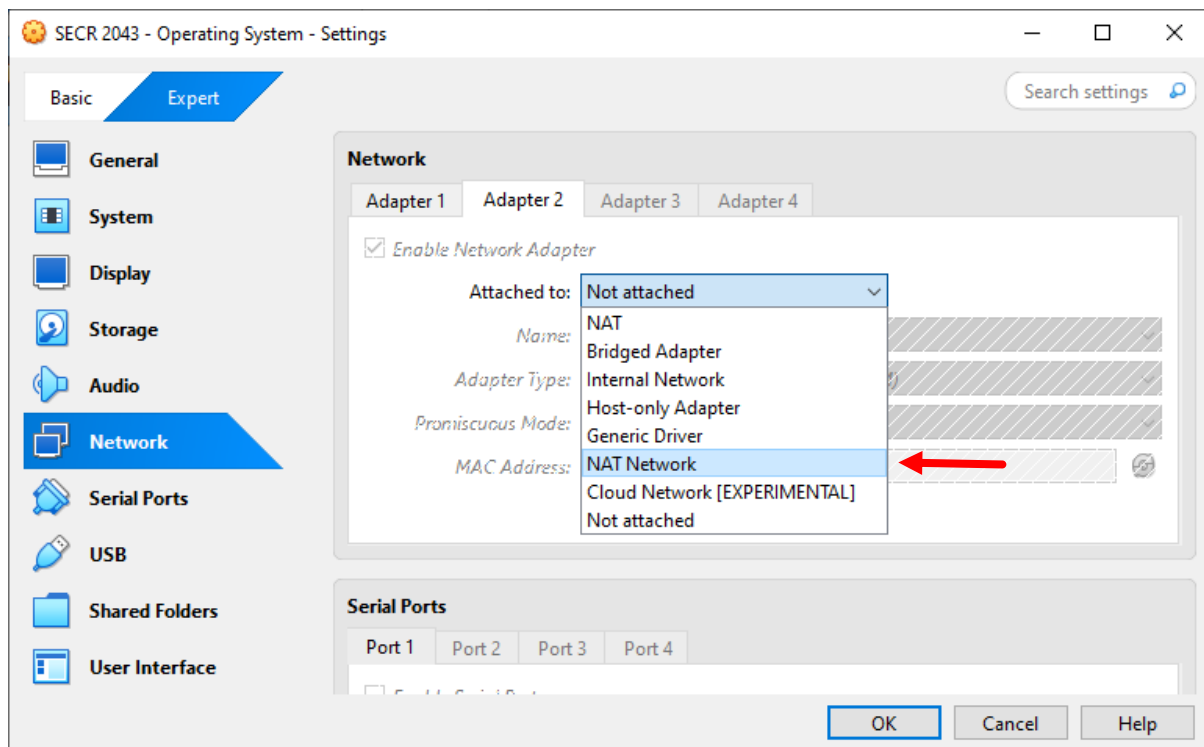


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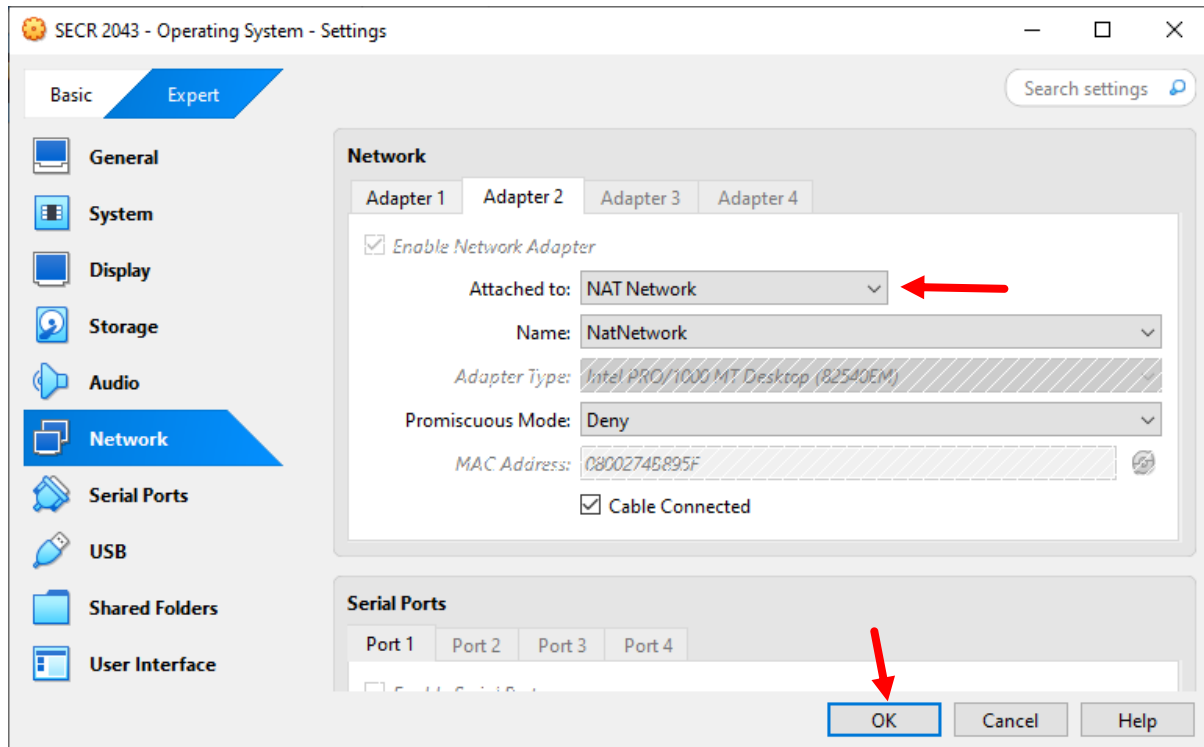
Next is to change the Attached to from **Not attached** to **NAT Network**.



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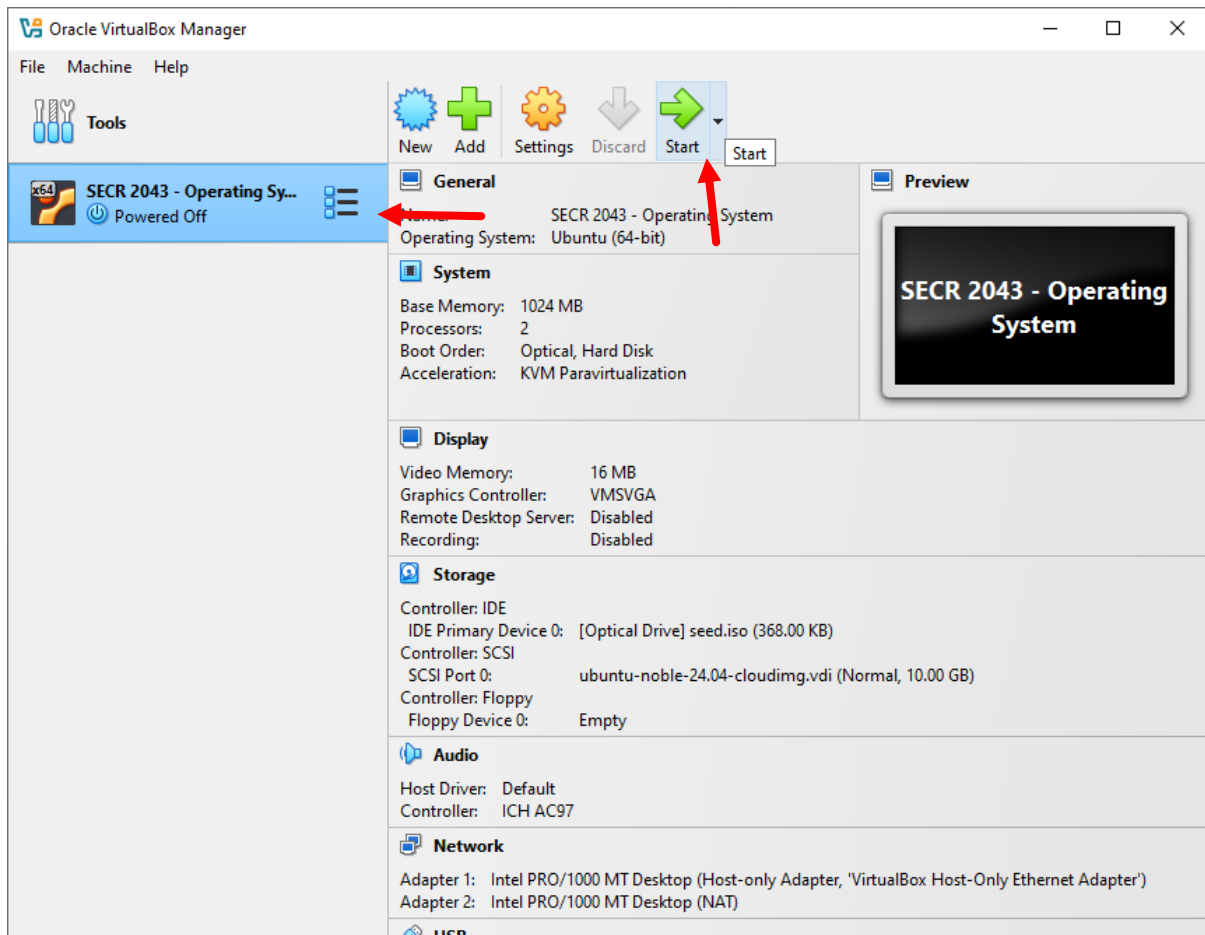
Don't forget to click **OK** button to confirm the changes.



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Start the virtual machine for the first time by selecting the created virtual machine and click Start icon.



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1.4 Post Installation

1.4.1 Login and Change password

Step 1: Login to the virtual machine

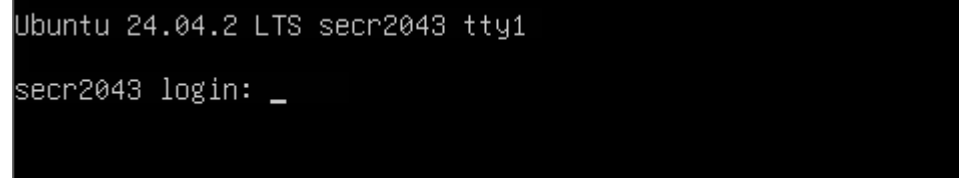
By default, the username is 'student' and the password is 'os123'.

Username = **student**

Password = **os123**

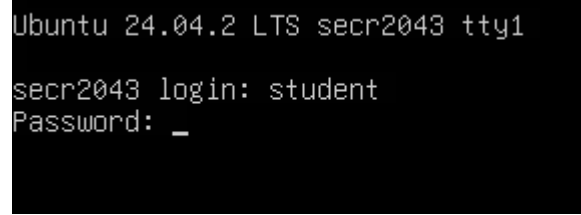
Enter the same to log in to your **Ubuntu 24.04 minimal Cloud Image** running on VirtualBox.

When the login screen appear like below. Just type the username next to the login prompt and enter.



```
Ubuntu 24.04.2 LTS secr2043 tty1
secr2043 login: _
```

Next, it will ask you to enter the password of the current user.



```
Ubuntu 24.04.2 LTS secr2043 tty1
secr2043 login: student
Password: _
```

When you type the password, it won't show on the screen. Don't worry, this is normal.

This is the classic security feature of any Unix or Unix like operating system when you accessing the machine remotely using ssh, by not displaying the typed password.

1.4.2 Add a new user with sudo access

Step 1: Well, we don't want to use default user, so create a **non-root user** with **sudo** rights, using the given commands:

```
sudo adduser muhammad
```

You can replace **muhammad** with whatever user name you want to give your new user.

Add created user to the sudo group.

```
sudo usermod -aG sudo muhammad
```

sudo is shortform of "super user do". Any user that is part of sudo group, has the authority of super user such as software installation, system configuration and deleting system files.

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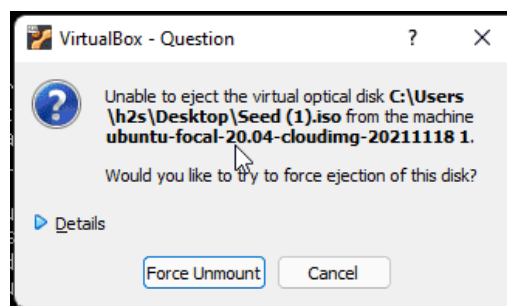
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1.4.3 Remove Seed ISO and restart your system

Step 1: Go to **Devices**-> Select **Optical Devices** and then **Remove disk from virtual drive**.



Step 2: After that click **Force Unmount** button.



Step 3: Shutdown the virtual machine

Now, we have removed the **seed.iso** file, simply shutdown your virtual machine system using this command:

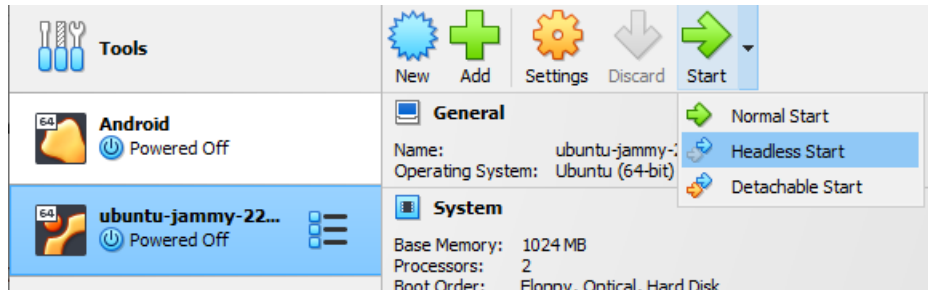
```
sudo halt -p
```

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2 Accessing Ubuntu 24.04 Virtual Machine Remotely

Step 1: Run Ubuntu Server 24.04 with headless start mode.



Step 2: Open Windows Terminal

Step 3: Run this command

```
ssh <username>@192.168.56.10
```

Use new username that you have created at section 1.4.2
e.g:-

```
ssh muhammad@192.168.56.10
```

The first time you use SSH, it will ask to add a public key for security. Just type **yes** to confirm..

As before, your password won't appear when you type it—this is a security feature.

```
~ $ ssh muhammad@192.168.56.10
The authenticity of host '192.168.56.10 (192.168.56.10)' can't be established.
ED25519 key fingerprint is SHA256:5Q819UGtKK2jgQPEBLtt1EZG2XqgX9+G30s9b2MesWI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.10' (ED25519) to the list of known hosts.
muhammad@192.168.56.10's password:
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

muhammad@secre2043:~$
```

Done.